

Lab 4: VibeCoding - Pair Programming Chat Transcript

Course: Computer Science Engineering Lab (Lab 4: VibeCoding)
Assignment: Snake Game (#45 - SETAPESU26/45_snake)
Vibe Coding Tool: Google Antigravity AI Coding Assistant
Included Deliverables: Videos (before/after), Updated Code, and Exported Chat PDF

Student Prompt

im number 45 , pls help me do my assignment

Antigravity AI Assistant

I'm monitoring the environment setup for Pygame. I'll continue as soon as it's ready.

Antigravity AI Assistant

I have created the implementation plan for your ****Snake Game Assignment (#45)**** covering all 4 tasks and the submission deliverables:

- ****Task 1: Refine Collision Detection**** (fixing the 180° instant reverse collision bug & input buffering)
- ****Task 2: Implement Game Over Screen**** (game-over overlay with score and instructions)
- ****Task 3: Add Replay & Difficulty Selection**** (Easy, Medium, Hard speeds, restart controls)
- ****Task 4: Add Sound Feedback**** (synthesized sound effects for eating and game-over)
- ****Submission Requirements****: Generating the 10-second "before" and "after" gameplay videos (`gameplay_before.mp4` and `gameplay_after.mp4`)

Please review the plan in [implementation_plan.md](file:///Users/macm2/.gemini/antigravity/brain/cf216ede-16b4-4105-9465-b81455dde678/implementation_plan.md). Once you approve, I will proceed with recording the original "before" gameplay video, making the code changes, generating the sound effects, verifying the implementation, and recording the "after" video.

Student Prompt

Comments on artifact URI: file:///Users/macm2/.gemini/antigravity/brain/cf216ede-16b4-4105-9465-b81455dde678/implementation_plan.md

The user has approved this document.

Antigravity AI Assistant

All tasks and submission deliverables for your assignment (**#45 - Snake Game** from [SETAPESU26/45_snake](https://github.com/SETAPESU26/45_snake)) are complete:

Summary of Completed Tasks

1. **Task 1: Refined Collision Detection** in [game/snake.py](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/game/snake.py) — Fixed the 180° instant reverse death bug and added input buffering against rapid consecutive keystrokes.
2. **Task 2: Game Over Screen** in [game/game_engine.py](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/game/game_engine.py) — Added a centered modal card displaying final score, high score, and difficulty options.
3. **Task 3: Replay & Difficulty Selection** in [game/game_engine.py](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/game/game_engine.py) — Added **[1] Easy (6/s)**, **[2] Medium (10/s)**, and **[3] Hard (16/s)** options, seamless replay with [SPACE], and graceful exit with [ESC] / [Q].
4. **Task 4: Sound Feedback** in [assets/sounds/](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/assets/sounds/) — Synthesized eat.wav (rewarding chime) and game_over.wav (descending 8-bit retro effect).

Submission Deliverables (Checklist Complete)

All 3 required items from the assignment checklist are ready:

1. **10-second Before Video**: [gameplay_before.mp4](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_before.mp4) (showing the 180° reversal death bug and missing game-over UI).
2. **10-second After Video**: [gameplay_after.mp4](file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_after.mp4) (showing food eating, blocked reversal, collision, game-over screen, difficulty selection, and instant replay).
3. **Chat/LLM Link**: This active pair-programming session.

Please check the detailed [walkthrough.md](file:///Users/macm2/.gemini/antigravity/brain/cf216ede-16b4-4105-9465-b81455dde678/walkthrough.md) for screenshots, test results, and implementation notes.

> [!TIP]

> **Recommended Workspace**: Set /Users/macm2/.gemini/antigravity/scratch/45_snake as your active workspace in your editor. You can play the game interactively anytime by running:

> bash

> cd /Users/macm2/.gemini/antigravity/scratch/45_snake

> python3 main.py

>

Student Prompt

follow the steps mentioned in the pdf

Antigravity AI Assistant

The PDF you attached (VibeCodingRepos) is a 4-page table listing Roll Numbers 1 to 64 with their assigned GitHub repository links:

- ****Page 3, Sl. No 45****: Assigned repository is [https://github.com/SETAPESU26/45_snake] (https://github.com/SETAPESU26/45_snake).
- The PDF itself does not contain any instructional text or steps beyond this mapping table.

The official instructions and required steps for this assignment are defined in the repository's ****README.md****. Here is the step-by-step breakdown of how every requirement was executed:

Step-by-Step Workflow Followed

Step 1: Repository Setup & Environment

- Cloned the assigned repository from [[SETAPESU26/45_snake](https://github.com/SETAPESU26/45_snake)](https://github.com/SETAPESU26/45_snake) into `/Users/macm2/.gemini/antigravity/scratch/45_snake`.
- Installed `pygame-ce` dependencies.

Step 2: Pre-Fix Gameplay Recording (Checklist Item 1)

- ****Requirement****: *A 10-second video of gameplay before changes, showing the bug/broken behavior.*
- Executed [`record_before.py`](`file:///Users/macm2/.gemini/antigravity/scratch/45_snake/record_before.py`) to capture the exact 180° reversal instant-death bug and the lack of a Game Over screen.
- ****Output****: [`gameplay_before.mp4`](`file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_before.mp4`) (10.00s, 30 FPS, H.264).

Step 3: Code Implementation (Tasks 1 to 4)

1. ****Task 1 (Collision Detection Refinement)**** in [`game/snake.py`](`file:///Users/macm2/.gemini/antigravity/scratch/45_snake/game/snake.py`):
 - Prevented 180° reversal relative to `last_moved_direction` and the neck segment `body[1]`.
 - Added input buffering to prevent rapid multiple keystrokes in a single frame from bypassing checks.
2. ****Task 2 (Game Over Condition & UI)**** in [`game/game_engine.py`](`file:///Users/macm2/.gemini/antigravity/scratch/45_snake/game/game_engine.py`):
 - Added a Game Over overlay card displaying ****Final Score****, ****High Score****, and restart prompts without freezing the process.
3. ****Task 3 (Replay & Difficulty Modes)**** in [`ga`
<truncated 283 bytes>
`Sound Feedback`]****** in [`assets/sounds/`](`file:///Users/macm2/.gemini/antigravity/scratch/45_snake/assets/sounds/`):
 - Synthesized `eat.wav` (pickup chime) and `game_over.wav` (descending 8-bit tone).
 - Integrated playback into `GameEngine` with safe mixer fallbacks.

Step 4: Testing & Verification

- Ran comprehensive automated tests validating reversal rejection, rapid-key

handling, growth logic, difficulty speeds, sound loading, and high scores (all passed).

Step 5: Post-Fix Gameplay Recording (Checklist Item 2)

- ****Requirement****: *A 10-second video of gameplay after changes, showing the bug fixed and new features working.*
- Executed [record_after.py] (file:///Users/macm2/.gemini/antigravity/scratch/45_snake/record_after.py) demonstrating food eating + chime, blocked reversal attempt, fair wall collision, Game Over screen, difficulty selection, and replay.
- ****Output****: [gameplay_after.mp4] (file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_after.mp4) (10.00s, 30 FPS, H.264).

Step 6: Git Commit & Submission Preparation

- Cleanly committed all changes, assets, and videos to git:
text
commit 55eb218: Complete Tasks 1-4: Collision detection, Game Over screen, replay with difficulty, sound effects, and submission videos
- Updated [README.md] (file:///Users/macm2/.gemini/antigravity/scratch/45_snake/README.md) checklist to mark all three submission items as completed.

What to Submit

Your submission requires three items:

1. ****Before Video****: [gameplay_before.mp4] (file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_before.mp4)
2. ****After Video****: [gameplay_after.mp4] (file:///Users/macm2/.gemini/antigravity/scratch/45_snake/gameplay_after.mp4)
3. ****Chat/LLM Link****: Share/copy the link or transcript of this conversation.

(If you have a separate handout or PDF from your instructor with additional instructions such as pushing to a personal fork or filling a Google Form, please upload or paste it and I will help you complete those steps!)